

I will attempt to use Kalman Filter approach to get the current ball position during the toss. The key concept is that the ball's current position is predicted using the motion model and camera corrects that.

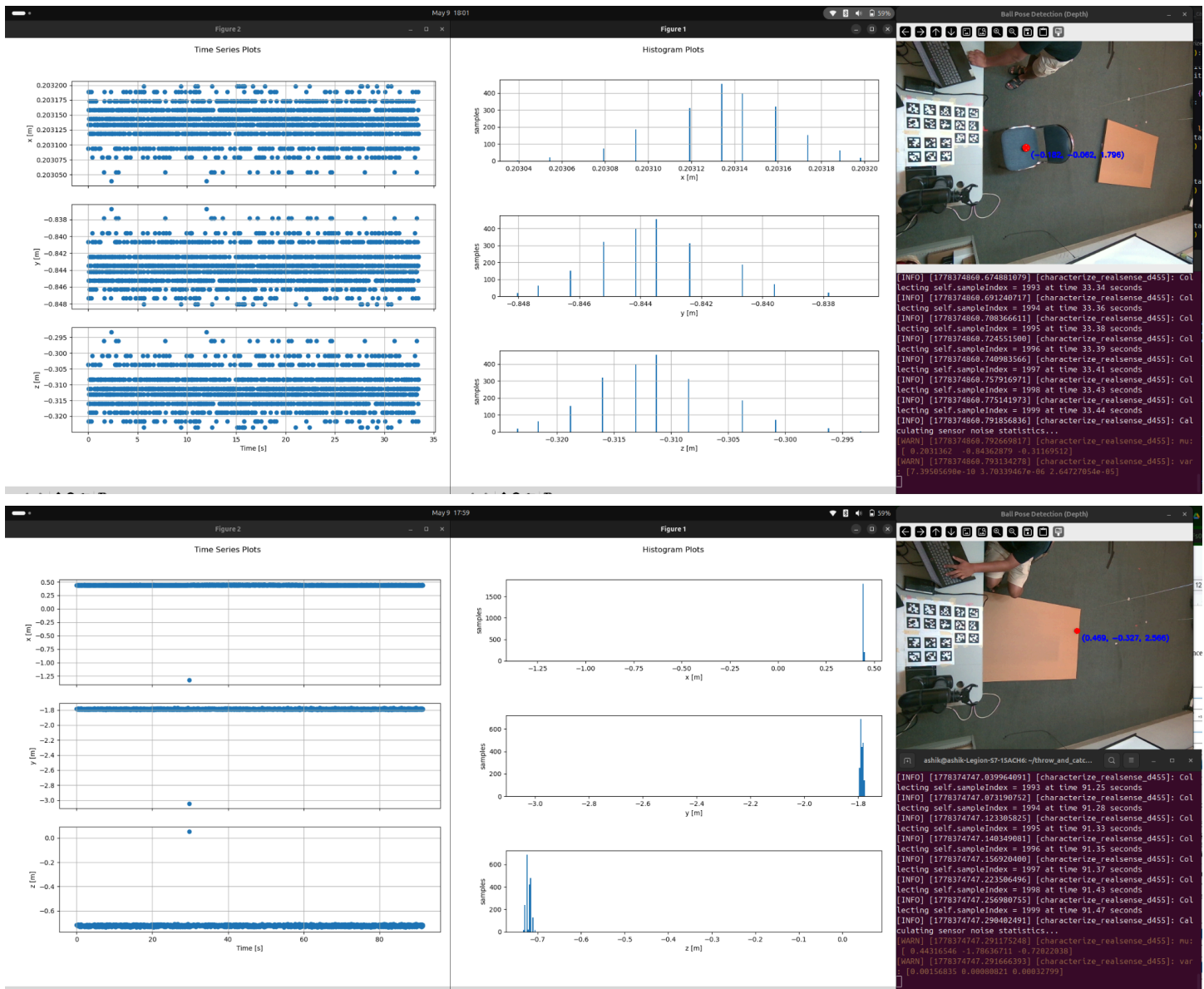
Characterize the Intel Realsense D455:

- Characterizing will be done in the 3d position in the robot frame. I.e. the msgs from /vision/ball_pose_robot. I will follow EE 470 lab 1.
- Created a pkg `characterize_sensor` and `characterize_rs` to do so.

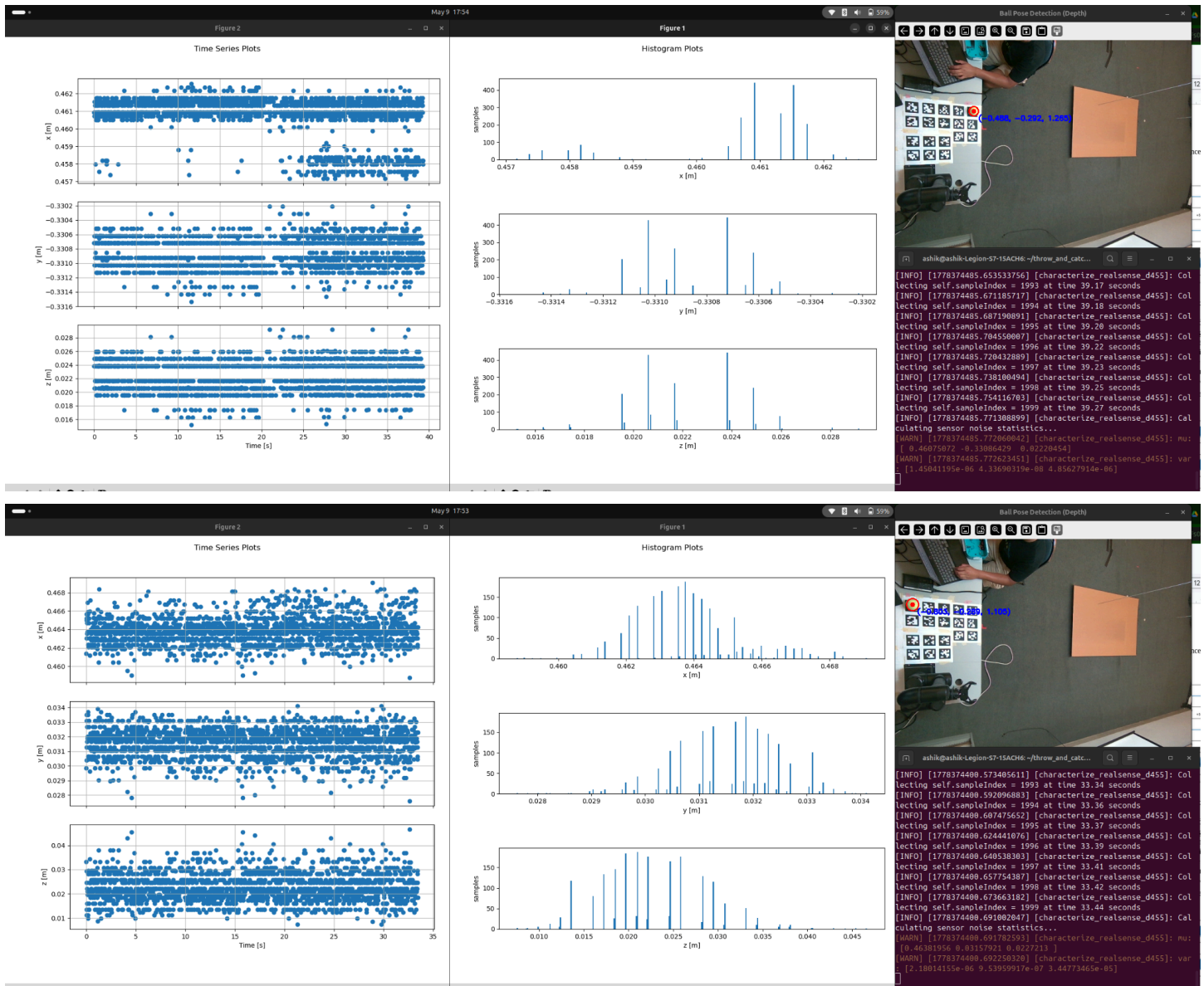
`characterize_rs`

Takes x amount of sample readings from vision/ball_pose_robot. Plots the time-series plot and histogram. And prints the mu and var for x, y, z.

Sample results of this node:



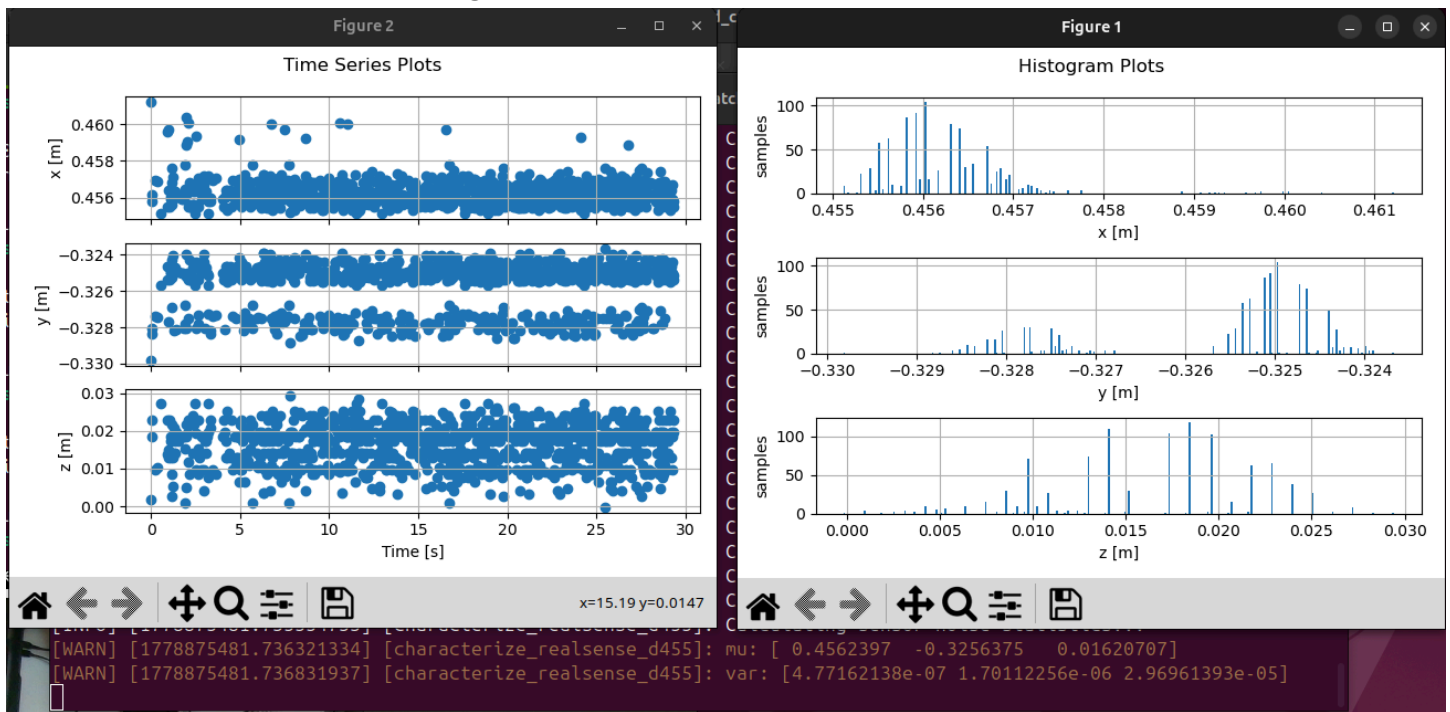
The outlier is likely a single false detection by yolo and can be ignored. I increased the required confidence level and such outliers did not appear again.



Chartization takeaways:

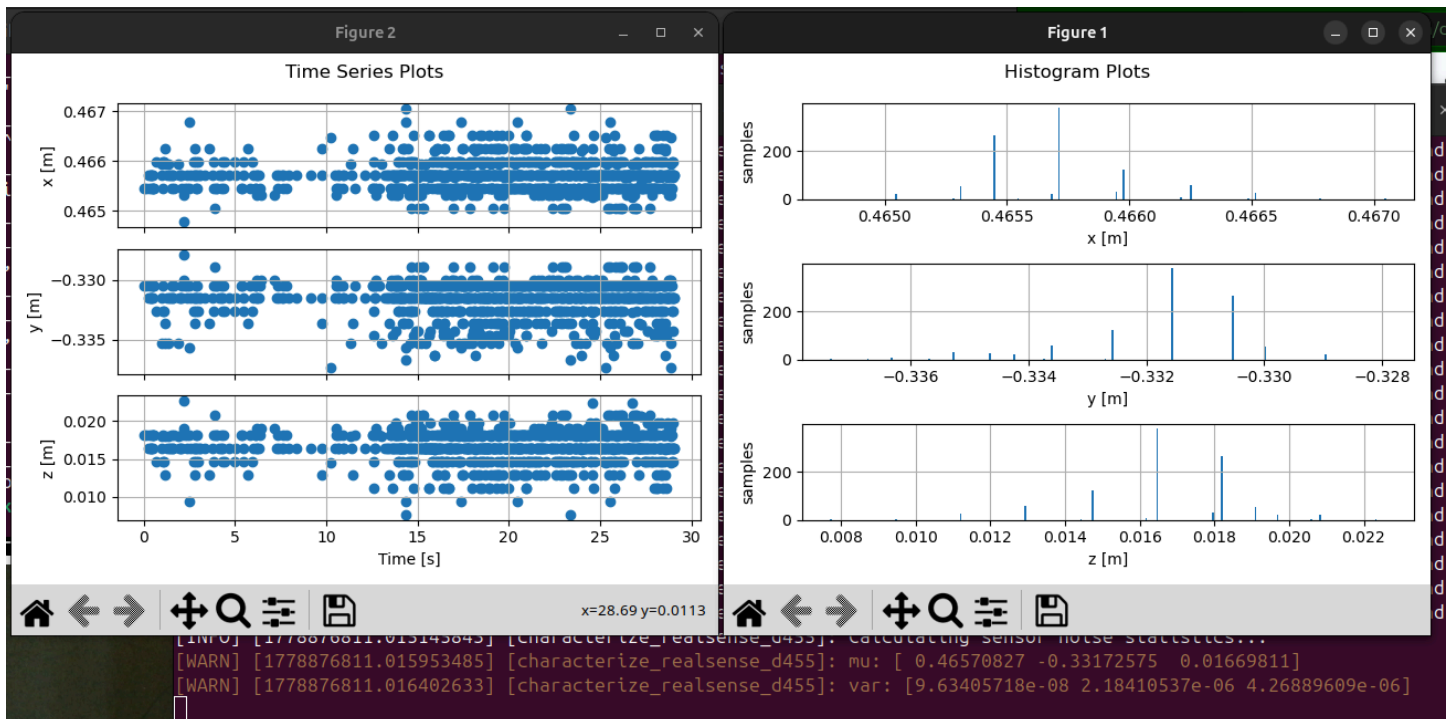
- The noise is gaussian like. However, it is quite different depending on where the ball is placed.
- Need to talk to farzan about how to merge such different mu and var into one R matrix.

Noise_characterization1 - ball on tag 13



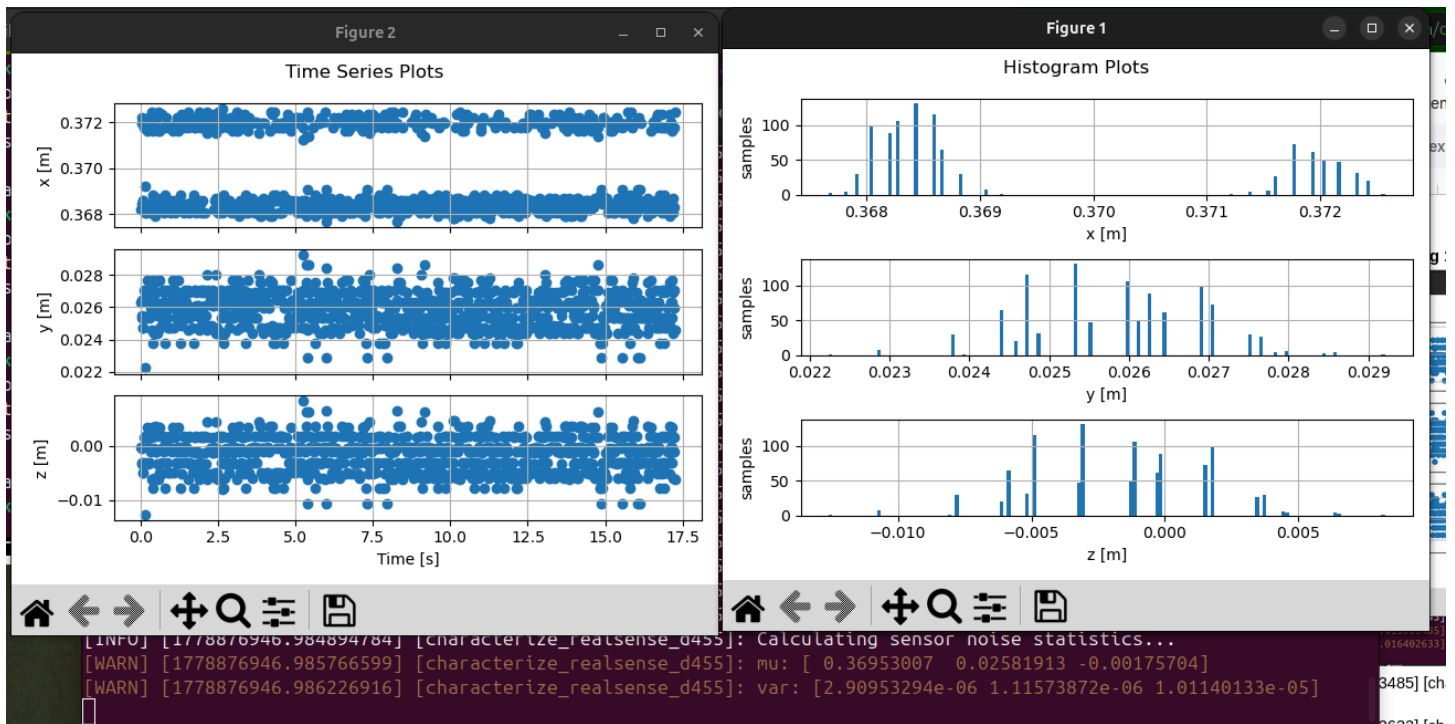
[WARN] [1778875481.736321334] [characterize_realsense_d455]: mu: [0.4562397 -0.3256375 0.01620707]
 [WARN] [1778875481.736831937] [characterize_realsense_d455]: var: [4.77162138e-07 1.70112256e-06 2.96961393e-05]

Noise_characterization2 - ball on tag 13

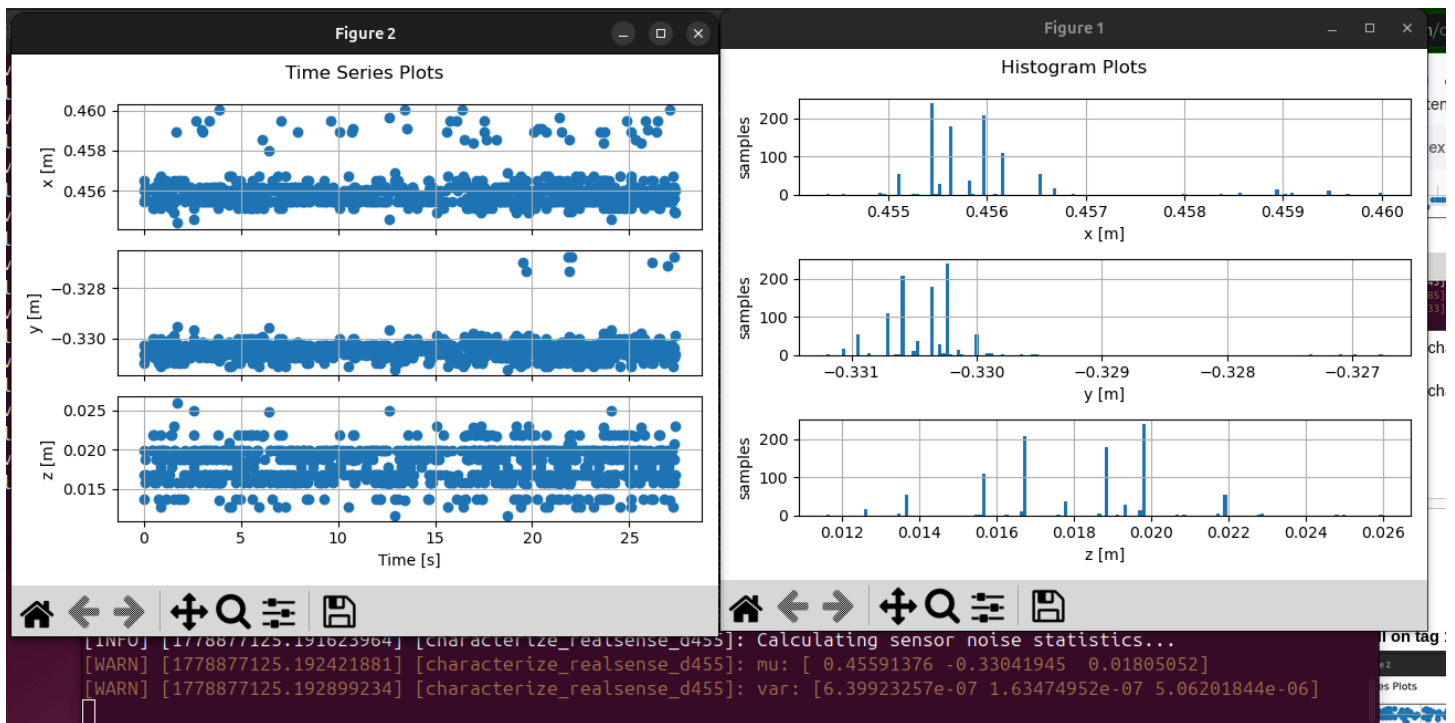


[WARN] [1778876811.015953485] [characterize_realsense_d455]: mu: [0.46570827 -0.33172575 0.01669811]
 [WARN] [1778876811.016402633] [characterize_realsense_d455]: var: [9.63405718e-08 2.18410537e-06 4.26889609e-06]

Noise_characterization3 - ball on tag 11



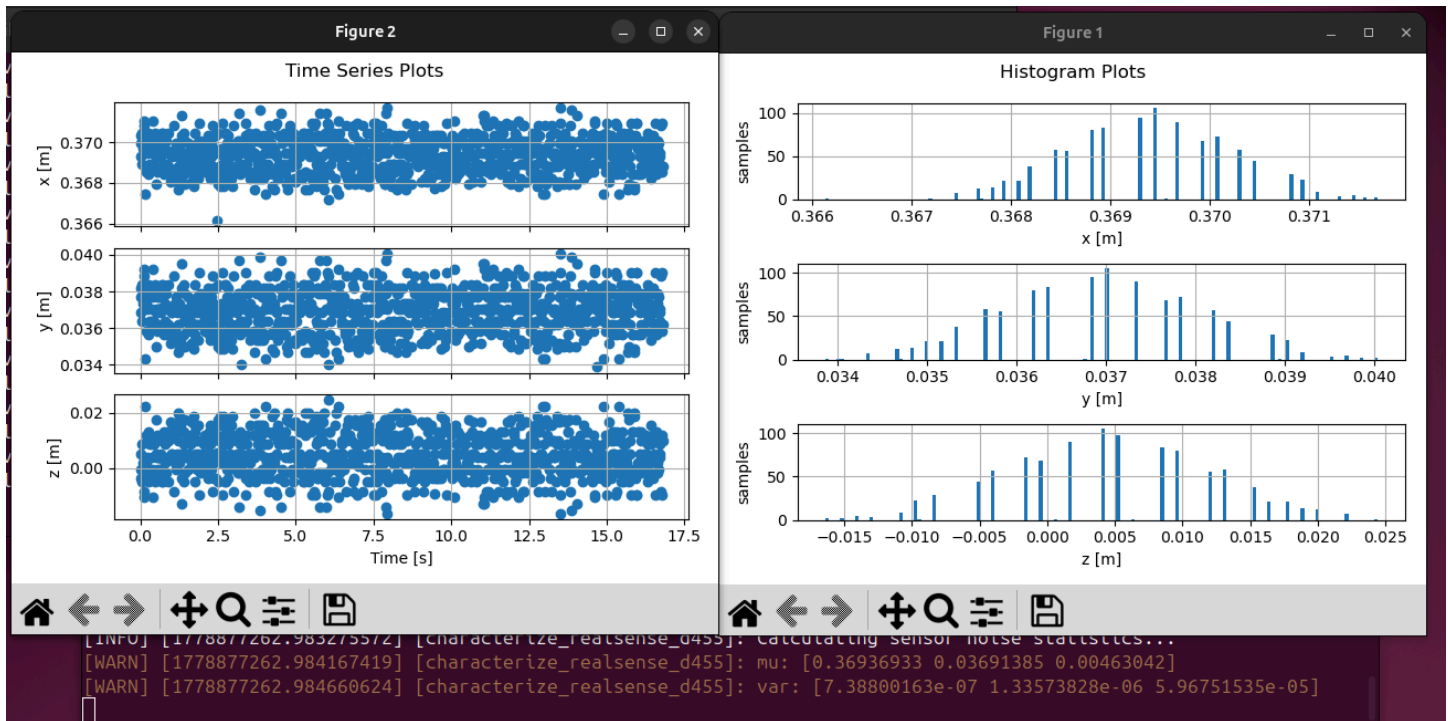
Noise_characterization4 - ball on tag 13



[WARN] [1778877125.192421881] [characterize_realsense_d455]: mu: [0.45591376 -0.33041945 0.01805052]

[WARN] [1778877125.192899234] [characterize_realsense_d455]: var: [6.39923257e-07 1.63474952e-07 5.06201844e-06]

Noise_characterization5 - ball on tag 11



```
[INFO] [1778877262.985275572] [characterize_realsense_d455]: calculating sensor noise statistics...
[WARN] [1778877262.984167419] [characterize_realsense_d455]: mu: [0.36936933 0.03691385 0.00463042]
[WARN] [1778877262.984660624] [characterize_realsense_d455]: var: [7.38800163e-07 1.33573828e-06 5.96751535e-05]
```

```
ros2 topic pub --once /arm_controller/joint_trajectory trajectory_msgs/msg/JointTrajectory "{
  joint_names: ['joint1','joint2','joint3','joint4','joint5','joint6'],
  points: [{
    positions: [-1.5708, 0.0, 1.5708, -1.5708, 1.50708, 0.0],
    time_from_start: {sec: 3}
  }]
}" # goes to home pose
```

```
ros2 topic pub --once /arm_controller/joint_trajectory trajectory_msgs/msg/JointTrajectory "{
  joint_names: ['joint1','joint2','joint3','joint4','joint5','joint6'],
  points: [{
    positions: [-0.5708, -0.50, 1.95708, -1.5708, 1.50708, 0.0],
    time_from_start: {sec: 3}
  }]
}" # goes to another pose
```

Velocity and position motion model

$$v(t + dt) = e^{-kdt}v(t) + \frac{1 - e^{-kdt}}{k}a$$

$$p(t + dt) = p(t) + \frac{1 - e^{-kdt}}{k}v(t) + \left(\frac{dt}{k} - \frac{1 - e^{-kdt}}{k^2}\right)a$$